SALES FORECASTING FOR RETAIL

Domain: Data Science

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B.Tech CSE

PROBLEM STATEMENT

- • Retail sales vary over time
- • Poor forecasting leads to:
- - Overstocking
- - Stock shortages

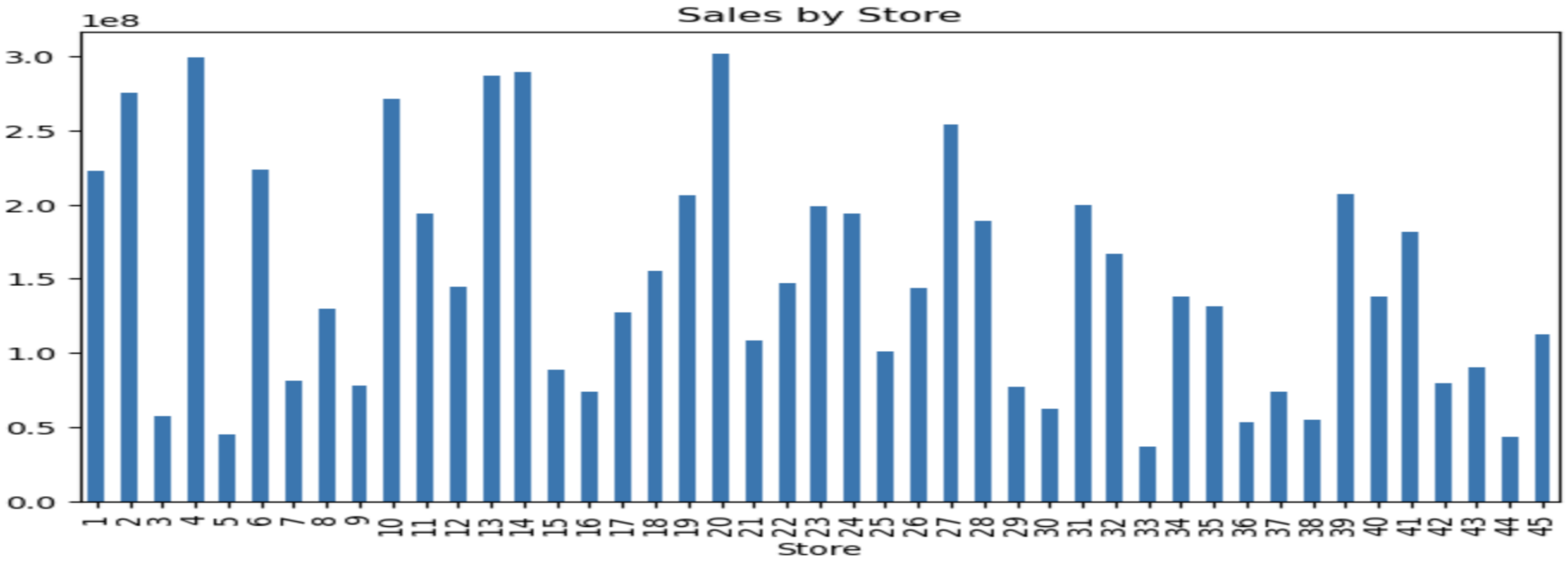
- Objective:
- Predict future sales using historical data

DATASET DESCRIPTION

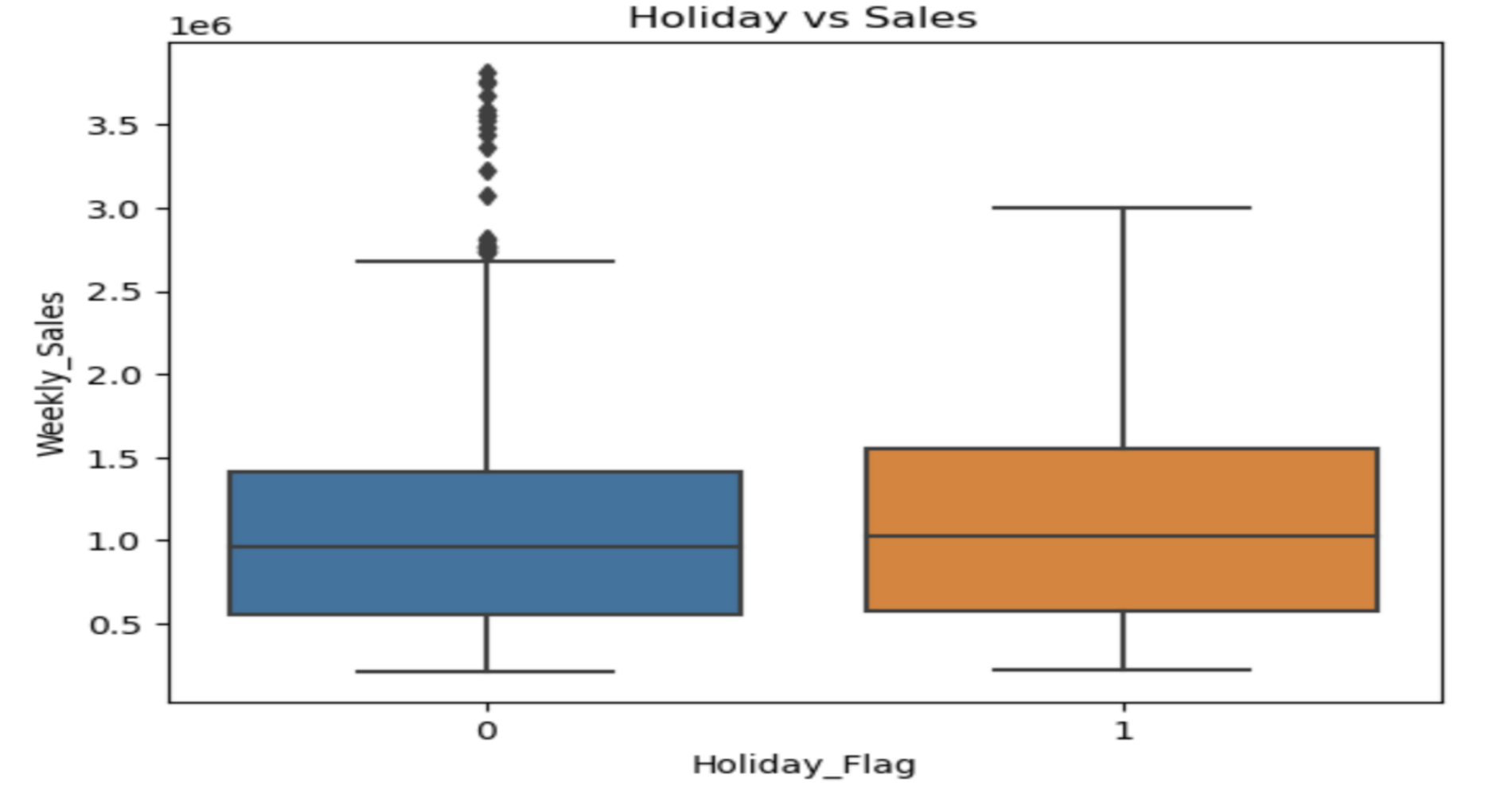
- • Walmart Sales Dataset
- • Features:
- - Date
- - Store
- - Weekly Sales
- - Holiday Flag
- - CPI, Unemployment

EXPLORATORY DATA ANALYSIS

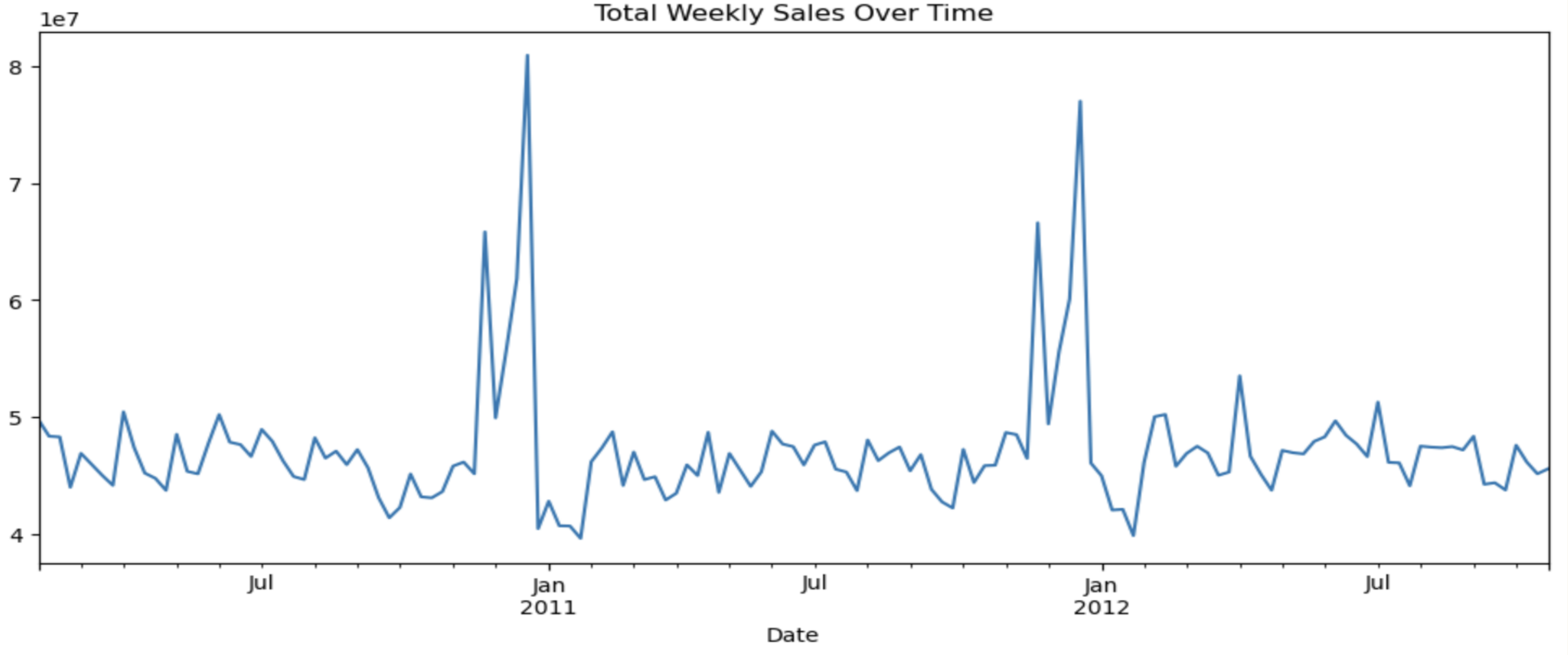
- Store-wise variation



- Holidays impact sales



- Seasonal trends observed



FEATURE ENGINEERING & MODEL USED

- Extracted:

  - Year, Month, Week

- Created lag feature:

  - Previous week sales

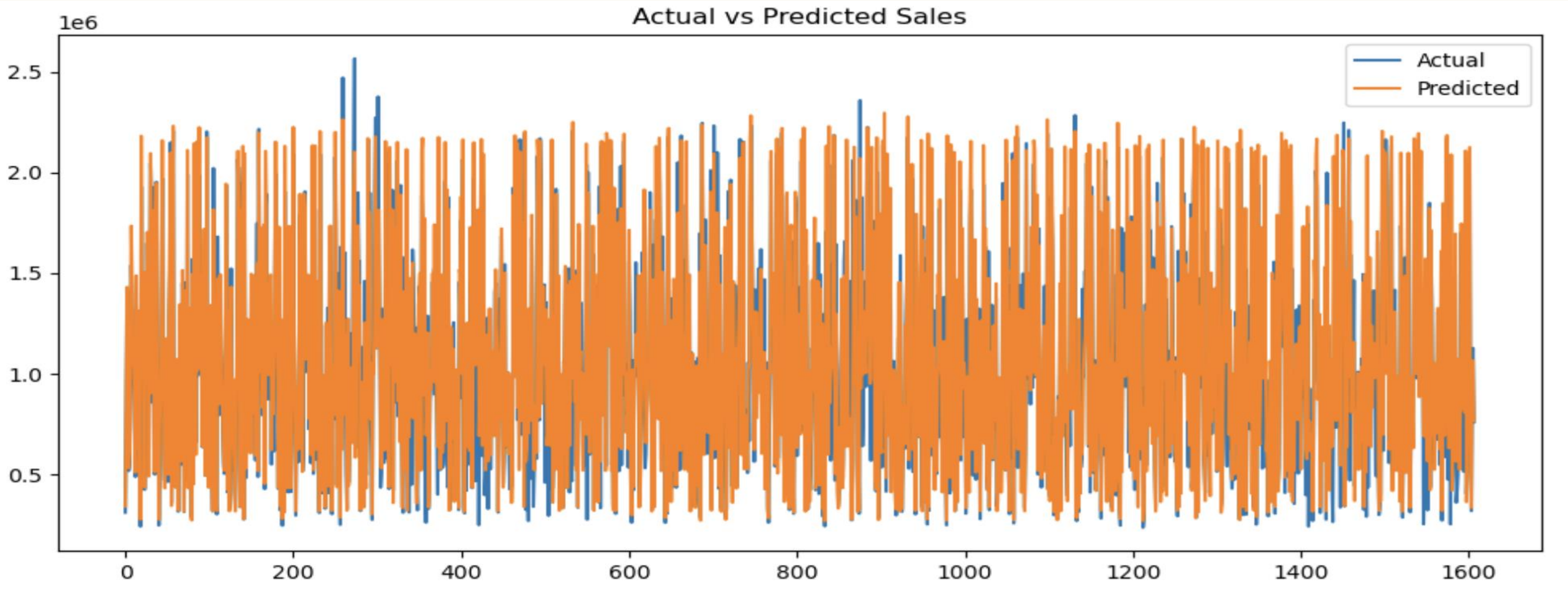
- Random Forest Regressor

  Reasons:

  - Handles complex patterns
  - Reduces overfitting
  - Works well with structured data

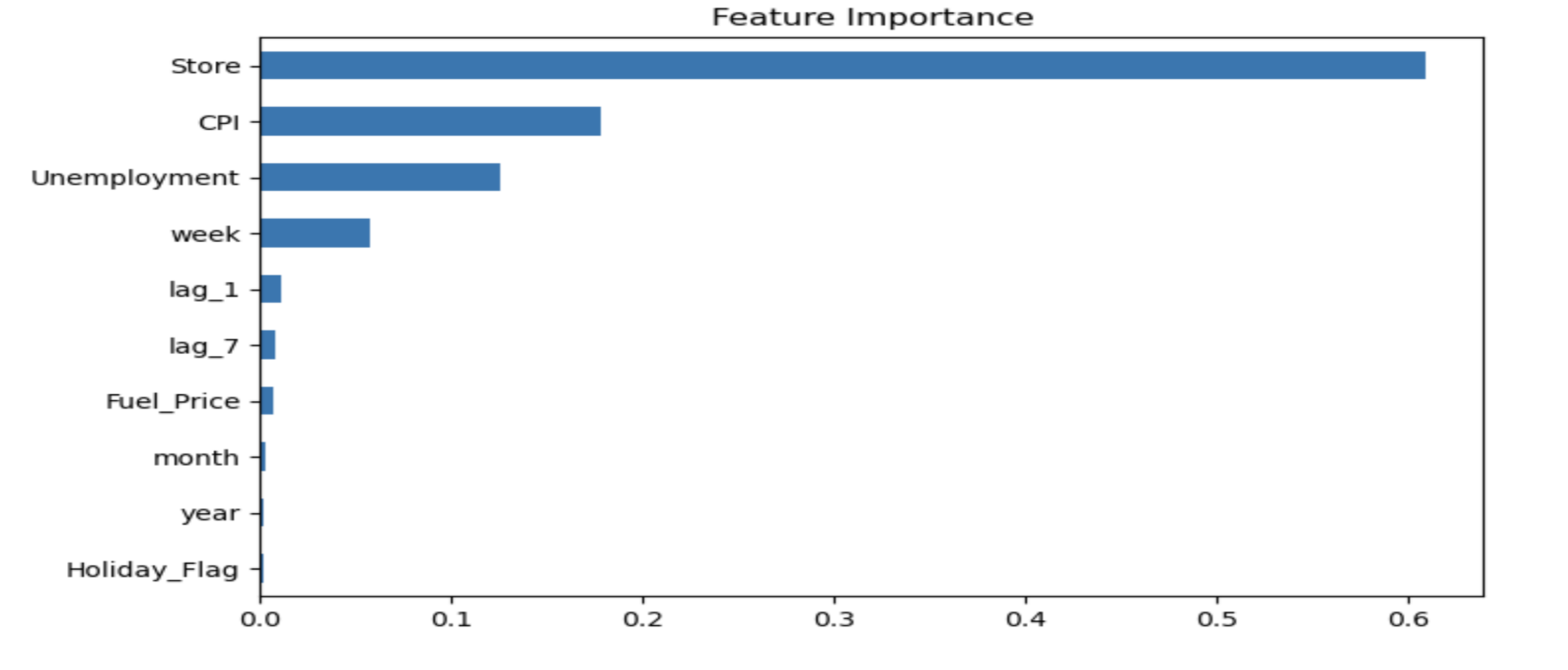
RESULTS & EVALUATION

- Metric: RMSE



FEATURE IMPORTANCE

- Store is most important feature



CONCLUSION

- Sales forecasting using ML is effective
- Store plays major role
- Model gives useful insights

FUTURE SCOPE

- Use advanced models (XGBoost, ARIMA)
- Add more features (promotions, weather)

THANK YOU